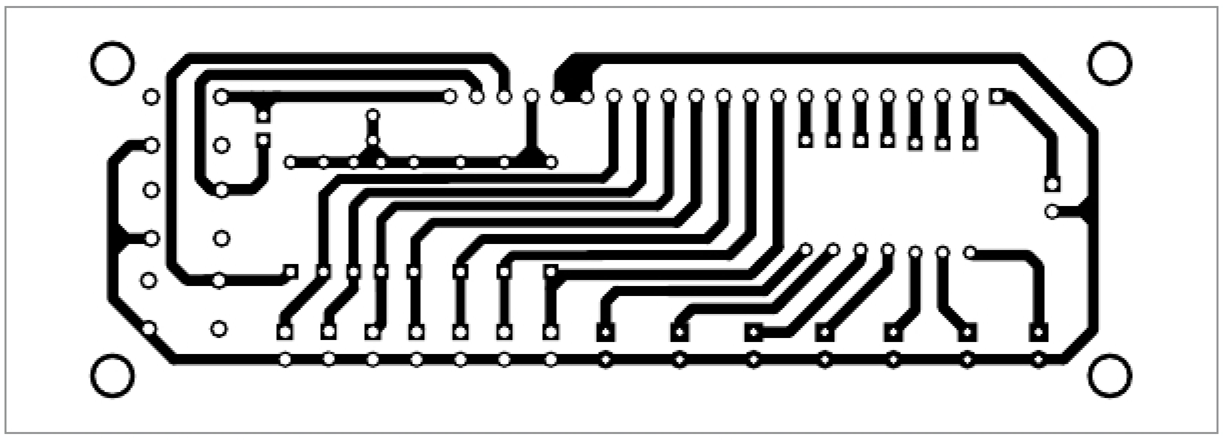

Fig. 3: Actual-size PCB layout of the window alarm annunciator

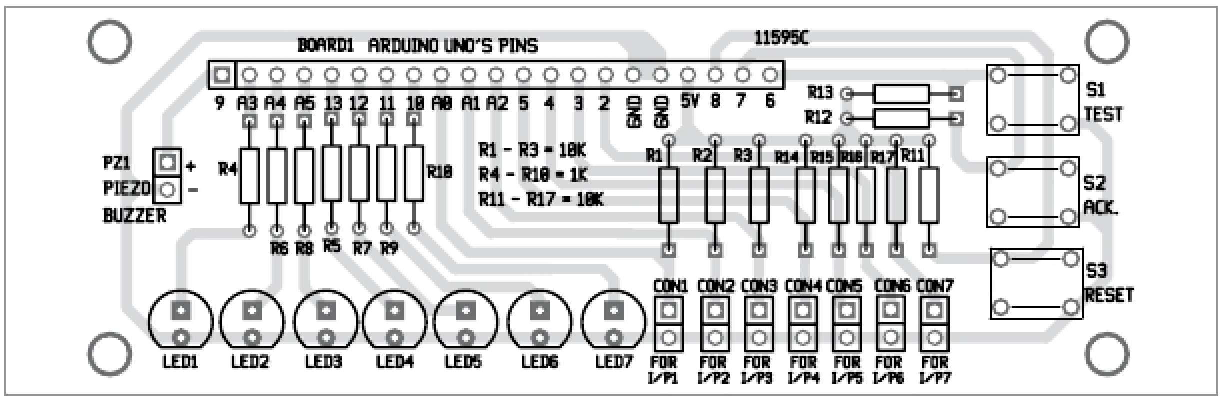

Fig. 4: Components layout for the PCB

PARTS LIST

Semiconductors:	
Board1	- Arduino Uno board
LED1-LED7	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1-R3, R11-R17	- 10-kilo-ohm
R4-R10	- 1-kilo-ohm
Miscellaneous:	
S1-S3	- Tactile switch
CON1-CON7	- 2-pin connector
	- Arduino Uno board with boot loader
	- Jumpers
	- 21-pin male Berg strip connector

by following the steps given below.

Sketch > Include libraries > Add. ZIP Library… > and browse the location where elapsed Millis file is saved. After adding the above library function, ensure that it appears in the list of library functions. Now, upload Annunciator.ino sketch file to the board.

Construction and testing

An actual-size PCB layout of the window alarm annunciator is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit, enclose it in a suitable box, along with Arduino Uno. Fix all LEDs on one side of the cabinet and piezo buzzer on the other. Power supply for the circuit is used through Arduino board and connected through the USB port of the laptop/desktop.

All inputs, I/P1 through I/P7, are potential-free external NO contacts with a common ground rail. That is, all inputs coming from the appliances/machineries should be NO. Input contacts should close when problems like overload or short-circuit occur in the load of one or more appliances/machineries.

For easy and quick testing, assemble the circuit on a breadboard, as per the given connection diagram. All inputs can be initiated, including acknowledge, test and reset inputs, by pressing the respective switches.

Caution. 1. Only potential-free external NO contacts should be used as alarm contacts (CON1 through CON7) with a common ground. Otherwise, external voltage injected into the circuit will damage Arduino Uno.

2. If long cables are used for alarm input, EMI suppression circuits/optocoupler circuits should be added to prevent spurious operation or damage. EFY

EFY Note
The source code of this project is available for free download at *source.efymag.com*


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